Project Design Phase-II
Data Flow Diagram & User Stories

Date	1 October 2026
Team ID	SWTID-2026-8643
Project Name	Script-Controlled ACL – Control Record Access Based on Field Value (ServiceNow)
Maximum Marks	4 Marks

Data Flow Diagrams:

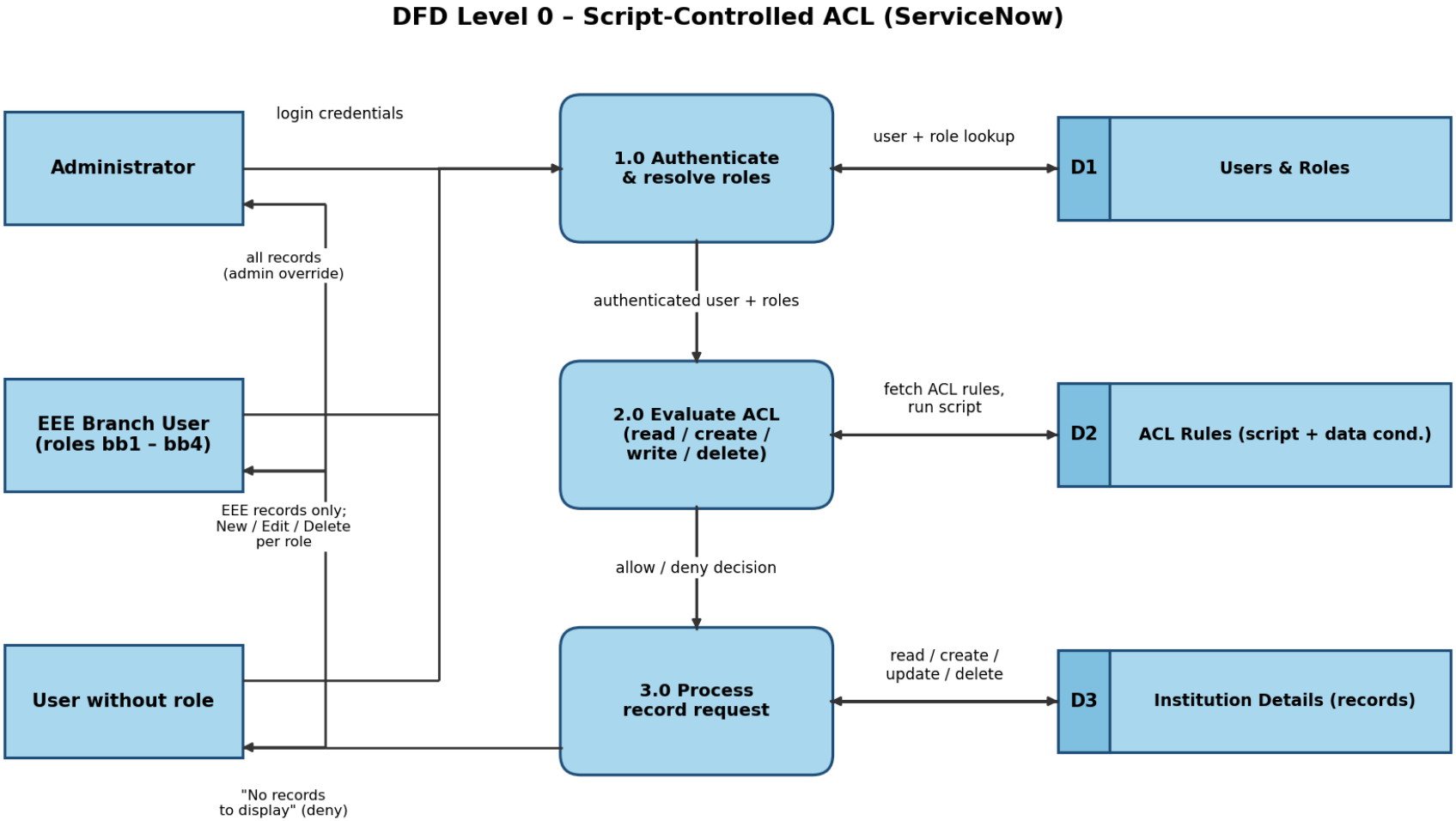
A Data Flow Diagram (DFD) visually represents how information moves through a system. It shows how data enters and leaves the system, what changes the information, and where data is stored.

For this project the system is the ServiceNow instance that protects the Institution Details table (u_institution_details). Each user request passes through role resolution and ACL evaluation before any record is read, created, edited or deleted.

Data flow description

Flow	Source	Destination	Data carried
F1	Administrator / EEE Branch User / User without role	1.0 Authenticate & resolve roles	Login credentials entered on the ServiceNow instance
F2	1.0 Authenticate & resolve roles	D1 Users & Roles	User record and assigned roles (admin, bb1, bb2, bb3, bb4) are looked up
F3	1.0 Authenticate & resolve roles	2.0 Evaluate ACL	Authenticated user together with the resolved role set
F4	2.0 Evaluate ACL	D2 ACL Rules	Read / Create / Write / Delete ACLs on u_institution_details are fetched; Requires-role, data condition (Branch is EEE) and ACL script are evaluated
F5	2.0 Evaluate ACL	3.0 Process record request	Allow / deny decision for the requested operation
F6	3.0 Process record request	D3 Institution Details	Records are read, created, updated or deleted only when the decision is allow
F7	3.0 Process record request	Administrator	All records regardless of branch (admin override)
F8	3.0 Process record request	EEE Branch User	Only EEE branch records; New button, inline edit and Delete appear according to roles bb2, bb3 and bb4
F9	3.0 Process record request	User without role	Empty list – "No records to display"

DFD Level 0


DFD Level 0 – Script-Controlled ACL (ServiceNow)
Administrator
EEE Branch User
(roles bb1 – bb4)
User without role
login credentials
all records
(admin override)
EEE records only;
New / Edit / Delete
per role
"No records
to display" (deny)
1.0 Authenticate
& resolve roles
2.0 Evaluate ACL
(read / create /
write / delete)
3.0 Process
record request
authenticated user + roles
allow / deny decision
user + role lookup
fetch ACL rules,
run script
read / create /
update / delete
D1
Users & Roles
D2
ACL Rules (script + data cond.)
D3
Institution Details (records)

User Stories

List of user stories for the product, grouped by functional requirement (epic).

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Administrator	User & Role Setup	USN-1	As an administrator, I can create the EEE User and the custom roles bb1, bb2, bb3 and bb4 and assign them to the user.	User and roles exist; all four roles are listed against EEE User	High	Sprint-1
Administrator	Table Setup	USN-2	As an administrator, I can create the Institution Details table (u_institution_details) with Student Roll Number, Student Name, Faculty Name, Branch, Email, Phone Number and Description fields.	Table is saved and all seven fields appear in the table columns	High	Sprint-1
Administrator	Table Setup	USN-3	As an administrator, I can add sample records with different Branch values (ECE, EEE, CSE).	List shows records for all three branches	Medium	Sprint-1
Administrator	Read ACL	USN-4	As an administrator, I can create a script-controlled read ACL on u_institution_details that requires role bb1 and a data condition of Branch is EEE.	ACL is active, Advanced is ticked, bb1 is in Requires role and the script is saved	High	Sprint-2
Administrator	Read ACL	USN-5	As an administrator, I can view all records regardless of branch.	On impersonating Admin, every record (ECE, EEE, CSE) is listed	High	Sprint-2
EEE Branch User (bb1)	Read ACL	USN-6	As an EEE branch user with role bb1, I can view only the EEE branch records.	List shows only records where Branch is EEE	High	Sprint-2
User without role	Read ACL	USN-7	As a user without the required role, I am prevented from viewing any records.	List shows "No records to display"	High	Sprint-2
Administrator	Create ACL	USN-8	As an administrator, I can create a create ACL on u_institution_details that requires role bb2.	ACL is active with bb2 in Requires role and no data condition	Medium	Sprint-3
Record Creator (bb1 + bb2)	Create ACL	USN-9	As a user with roles bb1 and bb2, I can see the EEE records and the New button to create a record.	EEE records are listed and the New button is visible	Medium	Sprint-3
Administrator	Write ACL	USN-10	As an administrator, I can create a write ACL on u_institution_details that requires role bb3.	ACL is active with bb3 in Requires role and no data condition	Medium	Sprint-3
Record Editor (bb1 + bb2 + bb3)	Write ACL	USN-11	As a user with roles bb1, bb2 and bb3, I can edit EEE branch records (for example, the Description field) from the list.	Inline edit saves the change, e.g. Description shows "ACL Is Executed"	Medium	Sprint-3
Administrator	Delete ACL	USN-12	As an administrator, I can create a delete ACL on u_institution_details that requires role bb4.	ACL is active with bb4 in Requires role and no data condition	Medium	Sprint-4
Record Manager (bb1 – bb4)	Delete ACL	USN-13	As a user with roles bb1, bb2, bb3 and bb4, I can view, create, edit and delete EEE branch records.	Selected EEE record is removed after Delete is confirmed	Medium	Sprint-4

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Administrator	Verification	USN-14	As an administrator, I can impersonate the EEE user, a user without roles and the admin to verify every ACL.	Each impersonated user sees exactly the access described in USN-5 to USN-13	High	Sprint-4